
ST-SIMDIFF: BALANCING SPATIOTEMPORAL SIMILARITY AND DIFFERENCE FOR EFFICIENT VIDEO UNDERSTANDING WITH MLLMs

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ABSTRACT

Multimodal Large Language Models (MLLMs) incur substantial computational overhead when applied to long-form videos, principally because effective video understanding conventionally requires the ingestion of a prohibitively large number of visual tokens. Existing efficiency-oriented methods primarily mitigate this burden by pruning or merging tokens according to either estimated importance or inter-token similarity; however, such strategies tend to underrepresent a crucial dimension of video semantics, namely content variation and transitional turning points, while also lacking a unified mechanism for modeling spatio-temporal dependencies. To address these limitations, we advance a complementary perspective: similarity should be exploited to identify redundancy, whereas difference should be preserved to capture salient events. On this basis, we introduce ST-SimDiff, a training-free framework that first constructs a spatio-temporal graph over visual tokens so that their heterogeneous spatial and temporal associations can be modeled in a common structure. We then perform a parallel dual-selection procedure: (1) similarity-based selection employs community detection to retain representative tokens that compactly encode stable information, and (2) temporal-difference-based selection localizes content-changing points so that tokens corresponding to key dynamic shifts are explicitly preserved. Through this joint treatment of persistent appearance and event-driven change, ST-SimDiff maintains both static and dynamic video content under a substantially reduced token budget. Extensive experiments demonstrate that our method markedly outperforms state-of-the-art alternatives while simultaneously reducing computational cost. Our code will be released upon acceptance.

1 INTRODUCTION

The rapid development of Large Language Models (LLMs) has substantially accelerated progress in Large Vision-Language Models (LVLMs), which now exhibit strong capabilities across image and video understanding tasks (Zhang et al., 2024c; Liu et al., 2024c). In prevailing video-processing pipelines, an LVLM first samples a video into a sequence of frames and then converts each frame into hundreds, or even thousands, of visual tokens that are subsequently provided to the LLM (Bai et al., 2025; Li et al., 2024). Although this paradigm has proven empirically effective, the resulting token count increases dramatically as video duration and resolution grow, thereby imposing severe computational and storage burdens that constrain the practical deployment of LVLMs in long-video analysis, real-time interaction, and other resource-sensitive scenarios (Fu et al., 2024b).

To mitigate this challenge, a variety of methods have been proposed to improve LVLM efficiency, and these approaches can be broadly organized into two categories. The first category comprises importance-based pruning methods. As observed by FastV (Chen et al., 2024), when LVLMs process visual information, attention distributions across different layers contain substantial redundancy; consequently, visual tokens receiving lower attention scores in deeper model layers may be pruned to reduce computational load. The second category consists of similarity-based merging or selection methods. Prior work has identified pronounced similarity among visual tokens, both across adjacent video frames and within individual frames: FrameFusion (Fu et al., 2024b) atten-

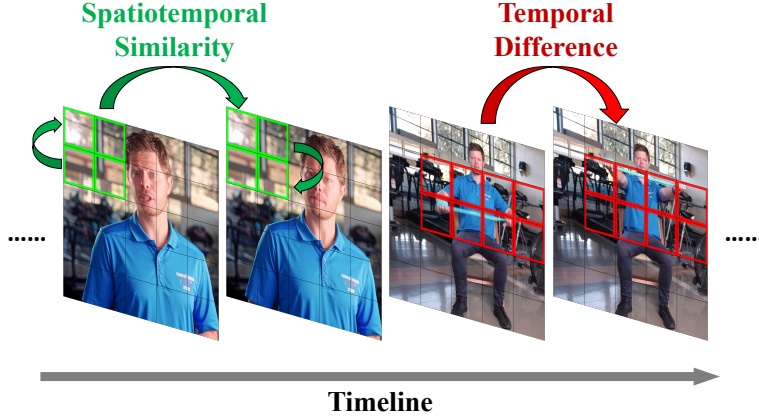


Figure 1: Core motivation of ST-SimDiff: efficient video understanding requires the simultaneous treatment of two complementary regimes. Spatiotemporal Similarity (left) enables the compression of redundant information both within each frame and across adjacent frames, whereas Temporal Difference (right) should be detected so that key actions and plot-defining events are retained.

uates redundancy by merging similar tokens from neighboring frames, whereas VisionZip (Yang et al., 2025) directly selects dominant tokens at the vision-encoder level.

Despite their demonstrated utility for visual-token reduction, existing methods exhibit two important limitations. First, token-pruning and token-merging strategies frequently analyze spatial correlations within a single frame or temporal correlations at corresponding positions across frames in isolation, rather than jointly modeling spatio-temporal relationships. This separation weakens their capacity to represent complex dynamic events whose semantics depend on coordinated spatial and temporal structure. Second, these methods share a more fundamental blind spot: by emphasizing informational commonality, such as similarity or importance, they tend to neglect abrupt differences and content changes. Because a video’s narrative is often organized around turning events, an algorithm that exclusively pursues similarity may smooth over sudden actions and thereby induce an incomplete or erroneous interpretation of the underlying content.

Motivated by this observation, we propose a new perspective: similarity should be used to identify redundancy, while difference should be used to capture key moments. An ideal video-token compression algorithm should therefore satisfy two requirements simultaneously: it should represent the stable content of a video with as few tokens as possible, and it should preserve critical changes with high precision. To this end, we introduce **ST-SimDiff**, a dual token-selection framework built on a spatio-temporal graph. The first component, **similarity-based representative token selection**, treats all visual tokens in a video as graph nodes and constructs edges according to feature similarity in the spatio-temporal domain. A graph community detection algorithm then identifies tightly connected token clusters, which correspond to stable and persistent visual elements, and a small number of high-centrality tokens from each cluster are selected as representatives. The second component, **difference-based event token selection**, focuses on temporal edges in the graph: when the similarity between corresponding tokens in adjacent frames drops sharply, we regard the transition as a turning point and retain the associated tokens as indispensable event evidence. Finally, the two selected token sets are merged and provided to the LLM. Our contributions can be summarized as follows:

- We are the first to argue that token similarity and token difference should receive comparable methodological attention in VLM-efficiency research.
- We develop a spatio-temporal graph framework that uniformly models complex relationships among video tokens, and we introduce a dual token-selection strategy capable of screening both representative tokens and pivotal event tokens.

- We conduct extensive experiments on video-understanding benchmarks, showing that ST-SimDiff substantially reduces the visual-token budget while maintaining, and in several cases improving, downstream performance.

2 RELATED WORK

2.1 LARGE VISION LANGUAGE MODELS

The rapid advancement of Large Language Models (LLMs) (Touvron et al., 2023; Achiam et al., 2023; Ouyang et al., 2022) has catalyzed substantial progress in multimodal understanding and has led to the emergence of numerous influential Large Vision Language Models (LVLMs) (Liu et al., 2024b; Bai et al., 2023). These LVLMs typically transform images or video frames into visual tokens through pre-trained visual encoders, after which the resulting tokens are injected into LLMs through alignment modules such as MLPs (Liu et al., 2024a;b) or Q-Formers (Li et al., 2023; Dai et al., 2023). This architectural pattern enables LVLMs to achieve strong performance on a wide range of multimodal tasks (Xi et al., 2025; Holdaway et al., 2024; Zhang et al., 2025a); in video understanding in particular, these models have shown increasing promise for processing longer, higher-resolution, and semantically more complex videos (Bai et al., 2025; Li et al., 2024; Zhang et al., 2024c; Liu et al., 2024c).

Nevertheless, the same representational capacity that makes LVLMs powerful also introduces substantial computational difficulty, especially in video-understanding scenarios. Because videos require the processing of continuous frame sequences, the number of visual tokens can grow far beyond that of static images, thereby creating a serious obstacle for both training and inference efficiency. Consequently, the efficient processing of massive video-token sequences, together with the reduction of their associated computational overhead, remains a central problem in contemporary LVLM research.

2.2 VISUAL TOKEN COMPRESSION

Motivated by the considerable overhead of video processing, visual-token compression has become an important route for improving the efficiency of Large Visual Language Models (LVLMs). Existing approaches primarily formulate compression through two complementary criteria: token importance and token similarity.

Early methods concentrated mainly on importance-based token pruning, seeking to identify and remove visual tokens that contribute comparatively little to model performance. Such methods typically rely on attention scores or related feature metrics to quantify token importance, as exemplified by FastV (Chen et al., 2024) and FasterVLM (Zhang et al., 2024b). Subsequent work (Shang et al., 2024; Ye et al., 2025) has increasingly incorporated token similarity, with the aim of reducing redundancy more effectively while retaining critical information. These methods recognize that even tokens deemed important may be highly similar to one another and thus informationally redundant. For example, FrameFusion (Fu et al., 2024b) combines similarity-based merging with importance-based pruning, VisionZip (Yang et al., 2025) reduces redundancy by selecting dominant tokens and merging contextual information, and VisPruner (Zhang et al., 2025b) leverages visual cues to eliminate redundant elements.

Despite these advances in visual-token redundancy reduction, existing work seldom addresses the intricate, global spatio-temporal similarity relationships that arise among video tokens. In video scenarios, redundancy is simultaneously spatial and temporal; however, similarity-driven methods such as FrameFusion primarily emphasize temporal similarity between adjacent frames and therefore underutilize richer spatio-temporal relational information. This limitation restricts their ability to exploit the inherent redundancy of video data, particularly under extreme token-reduction budgets where the omission of structurally informative relationships can produce larger performance degradation.

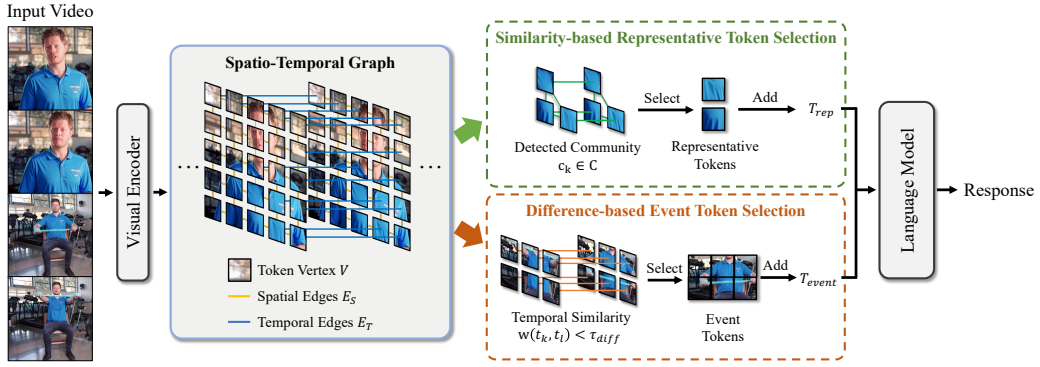


Figure 2: Overview of the ST-SimDiff framework, which comprises three interdependent components: Spatio-Temporal Graph Construction, Similarity-based Representative Token Selection, and Difference-based Event Token Selection.

3 PROBLEM DEFINITION

Given a video V and a text query Q , a standard Large Vision-Language Model (LVLM) first encodes the video as a complete sequence of N visual tokens, denoted by $T_{\text{full}} = \{t_1, t_2, \dots, t_N\}$. This encoding process is computationally expensive because the complexity of the self-attention mechanism grows quadratically with the token count N , i.e., $O(N^2)$. Our objective is therefore to design an efficient token-selection function, $f(\cdot)$, which receives the full token sequence T_{full} together with a compression ratio r and produces a substantially smaller subset $T_{\text{sub}} = f(T_{\text{full}}, r)$. The desired subset should maximize downstream LVLM performance while satisfying the budget constraint that its cardinality $|T_{\text{sub}}|$ remains approximately $r \cdot N$. The central difficulty, consequently, lies in constructing $f(\cdot)$ so that the reduced subset T_{sub} preserves the most task-relevant visual information despite aggressive compression.

4 ST-SIMDIFF

4.1 METHOD OVERVIEW

As illustrated in Figure 2, ST-SimDiff consists of two principal components that operate in parallel on a spatio-temporal graph constructed from the visual tokens of the input video: Similarity-based Representative Token Selection (SRTS) and Difference-based Event Token Selection (DETS). In the SRTS branch, we identify and condense stable, redundant content by applying community detection to the graph, thereby grouping highly similar tokens into clusters that typically correspond to persistent visual elements such as static backgrounds. Within each cluster, a small number of high-centrality tokens are selected to form the representative set T_{rep} . In the DETS branch, we instead target the video’s crucial dynamic transitions by examining temporal graph edges and detecting positions where the similarity between corresponding tokens in adjacent frames decreases sharply. Such high-difference locations indicate turning points, and the tokens associated with these transitions are retained as critical event tokens, forming the set T_{trans} . By combining these complementary branches, ST-SimDiff yields a final subset $T_{\text{sub}} = T_{\text{rep}} \cup T_{\text{trans}}$ that is simultaneously compact and information-rich, preserving persistent scene content while explicitly retaining key dynamic events.

4.2 SPATIO-TEMPORAL GRAPH CONSTRUCTION

Given a video, the vision encoder transforms it into a token sequence $T = \{t_1, t_2, \dots, t_N\}$. Each token $t_k \in T$ corresponds to a particular spatio-temporal location and is associated with a feature vector $\mathbf{x}_k$. We specify the location of t_k through its frame index $T(t_k)$, spatial height index $H(t_k)$, and spatial width index $W(t_k)$.

We represent the relationships among these tokens with a spatio-temporal graph $G = (V, E)$, where the vertex set V contains all tokens in T . The edge set E is defined as the union of two disjoint rela-

tional subsets, $E = E_S \cup E_T$, which respectively encode spatial adjacency and temporal continuity. The spatial edges E_S connect tokens that occupy adjacent spatial positions within the same frame:

$$E_S = \{(v_i, v_j) \in V \times V \mid T(t_i) = T(t_j) \text{ and } |H(t_i) - H(t_j)| + |W(t_i) - W(t_j)| = 1\} \quad (1)$$

The temporal edges E_T connect tokens that share the same spatial position across adjacent frames, thereby capturing local temporal continuity:

$$E_T = \{(v_i, v_j) \in V \times V \mid H(t_i) = H(t_j), W(t_i) = W(t_j), \text{ and } |T(t_i) - T(t_j)| = 1\} \quad (2)$$

For any edge $w(v_i, v_j) \in E$, the edge weight is defined as the cosine similarity between the feature vectors of the corresponding tokens:

$$w(v_i, v_j) = \frac{\mathbf{x}_i \cdot \mathbf{x}_j}{\|\mathbf{x}_i\| \|\mathbf{x}_j\|} \quad (3)$$

This sparse graph construction compactly encodes local spatial relationships together with temporal continuity, and thus provides a unified structural substrate for the subsequent dual-path token-selection procedure.

4.3 SIMILARITY-BASED REPRESENTATIVE TOKEN SELECTION

Video content is intrinsically characterized by substantial spatio-temporal redundancy. In a static scene that persists for several seconds, for example, background tokens remain highly similar over time; likewise, when an object moves across the screen, the tokens corresponding to that object may preserve strong semantic correlation even as their spatial positions change. Existing methods often treat temporal and spatial similarities separately, whereas our spatio-temporal graph is designed to capture their joint structure. Within this graph, semantically related tokens form dense ‘‘communities’’ or ‘‘clusters’’ according to feature similarity, regardless of their specific frame index or spatial location. Selecting representative tokens from each community therefore becomes an efficient mechanism for compressing redundant information while maintaining the core semantics of the video.

To identify strongly correlated token clusters, we first threshold the graph G . Specifically, we set a similarity threshold τ_{sim} and retain only those edges whose weights exceed this threshold, obtaining a new graph $G' = (V, E')$, where $E' = \{(v_i, v_j) \in E \mid w(v_i, v_j) > \tau_{\text{sim}}\}$. We then apply a graph community detection algorithm (e.g., the Louvain method (Blondel et al., 2008)) to G' to obtain token clusters $C = \{c_1, c_2, \dots, c_m\}$. For each community $c_k \in C$, its internal tokens are ranked and filtered according to centrality. The centrality score $S_c(t_a)$ of a token $t_a \in c_k$ is defined as the average similarity between t_a and all other tokens in the same community:

$$S_c(t_a) = \frac{1}{|c_k| - 1} \sum_{t_b \in c_k, b \neq a} w(t_a, t_b) \quad (4)$$

Subsequently, we set the intra-community retention rate to the globally specified compression ratio r and preserve the top $\lceil |c_k| \cdot r \rceil$ tokens with the highest centrality scores in each community c_k . This uniform filtering rule naturally accommodates all communities, including single-node communities composed of unique tokens, in which case the sole node is retained. Formally, the representative-token set T_{rep} is defined as:

$$T_{\text{rep}} = \bigcup_{k=1}^m \text{TopK}_{t \in c_k, \text{score} = S_c(t)}(\lceil |c_k| \cdot r \rceil) \quad (5)$$

where the TopK function returns the set of k elements with the highest scores from a given set. In this way, the strategy compresses static and persistent video content by retaining the most central tokens within each semantic cluster.

4.4 DIFFERENCE-BASED EVENT TOKEN SELECTION

If similarity characterizes the “norm” of a video, then difference characterizes its “events”. A model that focuses exclusively on similarity may be effective at recognizing “what is” present, yet remain insufficiently sensitive to “what happened” over time. Because video narratives are typically driven by key events, and because the essence of an event is change—the appearance of a new object, the initiation of an action, or a scene transition—these changes emerge in our formulation as sharp feature differences between temporally adjacent tokens. Accurately identifying such high-difference turning points is therefore essential for modeling dynamic video content and for answering temporally grounded questions such as “when” and “why”.

Temporal difference often signals the occurrence of a salient event: the entrance of a new object, a scene change, or the beginning or termination of an action can all produce pronounced feature variation at corresponding positions in adjacent frames. Accordingly, we analyze the temporal edges (E_T) in the spatio-temporal graph. For any temporal edge $(v_k, v_l) \in E_T$ connecting two temporally adjacent tokens t_k and t_l , we set a dynamic threshold τ (e.g., the 95th percentile of all temporal edge difference scores). When the difference score of (v_k, v_l) exceeds this threshold, equivalently when $w(t_k, t_l) < \tau_{\text{diff}}$, the subsequent token t_l is regarded as a critical event token and retained. Formally, the event-token set T_{event} is defined as:

$$T_{\text{event}} = \{t_l \mid \exists t_k \text{ s.t. } (v_k, v_l) \in E_T, \\ T(t_l) > T(t_k), \text{ and } w(v_k, v_l) < \tau_{\text{diff}}\} \quad (6)$$

This strategy preserves tokens that mark substantial temporal changes. By retaining these transitions, the compression process avoids discarding dynamic aspects of the video’s narrative that may be crucial for downstream understanding.

4.5 OVERALL REDUCTION PROCESS

In the full framework, we compute the representative-token set T_{rep} and the event-token set T_{event} in parallel, and we use their union, $T_{\text{candidate}} = T_{\text{rep}} \cup T_{\text{event}}$, as the initial candidate set. To satisfy the target token count $N_{\text{target}} = \lceil r \cdot N \rceil$ precisely, we further introduce a final pruning step: if $|T_{\text{candidate}}|$ exceeds N_{target} , we remove the $|T_{\text{candidate}}| - N_{\text{target}}$ tokens with the lowest importance. Following prior work (Chen et al., 2024), token importance is determined by the attention score assigned in the shallow layers of the LLM. This final adjustment enables the method to satisfy arbitrary computational budgets while preserving the tokens most likely to carry critical information.

Overall, the proposed reduction procedure combines graph-based token selection with attention-based dynamic pruning. The former exploits the intrinsic structure of video content, including both similarity and difference, to retain semantically central and event-sensitive information, whereas the latter supplies a flexible mechanism for matching a prescribed token budget. This design makes the compression process both principled and adaptive, yielding a robust balance between efficiency and performance.

4.6 COMPUTATIONAL COMPLEXITY ANALYSIS

The computational overhead of ST-SimDiff can be decomposed into three primary stages: Spatio-Temporal Graph Construction, Similarity-based Representative Token Selection (SRTS), and Difference-based Event Token Selection (DETS).

The first stage consists of graph construction. For each of the N visual tokens, we compute cosine similarity with its spatially and temporally adjacent neighbors. Because each token has only a small, constant number of neighbors, this stage requires a single traversal of the token sequence and has complexity $O(Nd)$, where d denotes the feature dimension of each token.

After graph construction, the SRTS module identifies token clusters. As described in our implementation (Section 5.2), we employ a connected-components algorithm, whose near-linear time complexity is $O(N + |E|)$. Since the edge count $|E|$ is at most $3N$, this term simplifies to $O(N)$. The subsequent within-community filtering step could, in principle, be more expensive: a naive all-pairs similarity computation inside each community c_k would require $O(|C| \cdot |c_k|^2 \cdot d)$. To prevent this component from dominating the overall runtime, we constrain the maximum size of each

Method	VideoMME				LongVideoBench			EgoSchema
	Short	Medium	Long	Overall	Relation	Perception	Overall	
Upper Bound (Full Performance)								
LLaVA-Video	-	-	-	63.3	-	-	58.2	57.3
Token Retain Ratio $r = 30\%$								
FastV	68.2	58.6	51.2	59.3	48.0	59.9	53.5	51.3
PruMerge	70.2	57.3	52.2	59.9	49.6	60.5	54.7	50.9
FasterVLM	71.3	57.6	51.4	60.1	51.0	61.3	55.8	52.6
VisionZip	68.4	55.9	50.4	58.3	46.9	60.3	53.2	53.0
FrameFusion	74.0	59.8	50.0	61.3	49.7	63.3	56.0	53.0
ST-SimDiff (Ours)	74.7	61.9	53.0	63.2	52.5	63.2	57.5	56.0
Token Retain Ratio $r = 50\%$								
FastV	73.9	61.4	51.3	62.2	50.1	62.2	55.7	54.7
PruMerge	72.3	58.7	52.8	61.3	51.4	63.2	56.9	54.6
FasterVLM	74.0	59.2	51.8	61.7	51.3	62.2	56.4	56.2
VisionZip	72.6	57.7	51.6	60.6	51.3	63.2	56.8	54.2
FrameFusion	74.6	61.2	52.0	62.6	51.7	64.2	57.6	55.8
ST-SimDiff (Ours)	76.2	62.4	52.7	63.8	52.3	64.3	57.9	57.3

Table 1: Performance comparison on long-form video-understanding benchmarks using LLaVA-Video-7B with 64 input frames (%). The strongest result among all methods is highlighted in **bold**.

community; in the rare case that a community exceeds a predefined threshold (e.g., $|c_k| > \sqrt{N}$), it is partitioned. Although graph clustering algorithms such as Louvain and Leiden provide more elaborate community definitions, their computational cost is higher, approaching $O(N \log N)$. We therefore adopt the simpler connected-components procedure to minimize processing overhead.

The DETS module performs a single pass over all temporal edges in the graph, identifying significant changes by comparing token similarities against the selected threshold. This process also has linear complexity $O(Nd)$.

In summary, the total complexity of ST-SimDiff is governed by linear-time operations and thus remains $O(Nd)$ overall. This cost is substantially lower than the quadratic complexity of the Large Language Model self-attention mechanism, $O(N^2d)$, and consequently the proposed method achieves meaningful token reduction with only a negligible computational footprint relative to the dominant inference cost.

5 EXPERIMENTS

5.1 EXPERIMENTAL SETUP

Baselines To evaluate the effectiveness of our method comprehensively, we adopt LLaVA-Video (Zhang et al., 2024c) and NVILA (Liu et al., 2024c) as base models and compare ST-SimDiff with a diverse set of state-of-the-art video-token compression techniques. The selected baselines instantiate multiple reduction paradigms, including **importance-based reduction methods** such as FastV (Chen et al., 2024) and FasterVLM (Zhang et al., 2024b), as well as **hybrid reduction methods** such as VisionZip (Yang et al., 2025), FrameFusion (Fu et al., 2024b), and PruMerge (Shang et al., 2024). To make the comparison both fair and diagnostically complete, we additionally report the original uncompressed model (Vanilla) as a performance upper bound and Random Sampling with an equivalent token count as a lower bound.

Method	VideoMME				LongVideoBench			EgoSchema
	Short	Medium	Long	Overall	Relation	Perception	Overall	
Upper Bound (Full Performance)								
NVILA-Video	-	-	-	61.5	-	-	56.3	52.9
Token Retain Ratio $r = 30\%$								
FastV	69.4	55.3	48.8	57.9	50.8	55.6	53.0	49.7
PruMerge	71.0	53.7	49.9	58.2	49.3	58.1	53.4	47.5
FasterVLM	72.6	56.9	51.0	60.1	48.7	57.8	53.0	49.3
VisionZip	71.4	56.2	49.7	59.1	46.3	56.2	50.9	48.9
FrameFusion	72.1	55.7	48.7	58.8	51.3	59.3	54.9	51.3
ST-SimDiff (Ours)	73.3	57.7	49.7	60.2	51.7	59.2	55.2	51.7
Token Retain Ratio $r = 50\%$								
FastV	71.9	58.3	49.3	58.9	49.5	58.8	53.9	50.2
PruMerge	71.2	54.8	46.8	57.6	49.7	58.7	53.9	48.9
FasterVLM	74.6	57.8	50.1	60.8	48.5	58.1	53.0	50.5
VisionZip	72.8	58.1	50.6	60.5	50.4	58.9	54.4	50.3
FrameFusion	72.7	57.2	48.3	59.4	51.0	59.2	54.8	52.6
ST-SimDiff (Ours)	73.9	59.7	51.4	61.7	52.3	61.3	56.5	52.5

Table 2: Performance comparison on long-form video-understanding benchmarks using NVILA-Video-8B with 64 input frames (%). The strongest result among all methods is highlighted in **bold**.

Benchmarks To assess long-video understanding ability, we employ three challenging benchmark datasets: VideoMME (Fu et al., 2024a), LongVideoBench (Wu et al., 2025), and EgoSchema (Mangalam et al., 2024). VideoMME provides a comprehensive evaluation suite for foundational video understanding, covering diverse video types and three duration categories: Short, Medium, and Long. LongVideoBench measures the capacity to retrieve and reason over fine-grained details in long videos through a “referring reasoning” task, with videos spanning four lengths: 15s, 60s, 600s, and 3600s. EgoSchema is a question-answering dataset centered on long-term, first-person videos, thereby testing whether a model can infer character intentions and causal chains of actions.

5.2 IMPLEMENTATION DETAILS

We conduct all experiments on NVIDIA L20 GPUs with 48GB VRAM under Ubuntu 22.04, and inference evaluation is implemented with the *lmms-eval* (Zhang et al., 2024a) library. For LLaVA-Video, we follow the original configuration of 64 input frames; to ensure that computational efficiency and resource consumption are compared under analogous conditions, we likewise set the NVILA input-frame count to 64. For the principal hyperparameters, we set the similarity threshold to $\tau_{\text{sim}} = 0.8$ and the difference threshold to $\tau_{\text{diff}} = 0.2$. Experiments are conducted at token compression rates of $r = 30\%$ and $r = 50\%$. In the practical implementation, after considering the trade-off between additional processing time and performance improvement, we use connected-component finding as the community detection algorithm.

5.3 COMPARISON WITH STATE-OF-THE-ARTS

To evaluate the proposed ST-SimDiff framework in a comprehensive manner, we compare it with a set of mainstream compression methods for efficient video language models (VLMs). All experiments are conducted on three widely used long-form video-understanding benchmarks, namely VideoMME, LongVideoBench, and EgoSchema. To examine the generalization ability and robustness of the method, we report detailed results for two distinct base models, LLaVA-Video-7B and

Method	VideoMME	LongVideoBench	EgoSchema
Token Retain Ratio $r = 30\%$			
Baseline	60.3	56.2	54.8
+ Sim	Spatial	61.5	56.5
	Temporal	61.7	56.8
	Spa. + Tem.	62.6	57.0
++ Diff (Proposed)	63.2	57.5	56.0
Token Retain Ratio $r = 50\%$			
Baseline	63.2	56.7	56.5
+ Sim	Spatial	63.3	57.0
	Temporal	63.7	57.2
	Spa. + Tem.	63.7	57.8
++ Diff (Proposed)	63.8	57.9	57.3

Table 3: Component-wise ablation results (%), quantifying the contributions of similarity selection and difference selection.

NVILA-8B, under token-retention ratios of 30% and 50%. The complete results are presented in Table 1 and Table 2.

(1) ST-SimDiff consistently outperforms all competing methods across the evaluated configurations. Particularly notably, at a 50% token-retention ratio, the overall performance of our method on both base models not only exceeds that of other compression algorithms, but on several benchmarks also matches or surpasses the original model that uses 100% of the tokens. This pattern substantiates the effectiveness and novelty of ST-SimDiff while motivating a closer examination of the strategies and limitations embodied by existing baselines.

(2) Among importance-based baselines (e.g., FastV and FasterVLM), FasterVLM typically delivers stronger performance. Nevertheless, these methods share a limitation in their inefficient treatment of the temporal redundancy that is pervasive in videos, since they tend to retain tokens that are important yet repetitive and thereby constrain the efficiency of information compression. Hybrid baselines (e.g., PruMerge, VisionZip, and FrameFusion) are generally more competitive than importance-only approaches, with FrameFusion usually achieving the strongest overall results. Even so, their common optimization objective remains the identification and preservation of representative “commonality” information, which creates a theoretical blind spot: key narrative evidence associated with “turning points” and “abrupt events” may be insufficiently preserved.

(3) In summary, existing baselines reflect a progressively deeper understanding of video “redundancy,” yet they continue to face two central challenges: first, they lack a synergistic analysis of joint spatio-temporal similarity among visual tokens, which makes complex redundancies difficult to capture; second, they generally neglect explicit “difference” detection and therefore do not actively preserve key changes that determine plot development. The performance gains of ST-SimDiff arise from addressing both issues simultaneously, through a spatio-temporal graph that uniformly models complex relationships and a difference-detection mechanism that explicitly protects event-defining transitions.

5.4 ABLATION STUDY

To validate the effectiveness of each core component in ST-SimDiff, we conduct a series of ablation studies that isolate the contribution of each selection mechanism. Starting from a Baseline strategy containing only fundamental importance-based pruning, we progressively introduce the proposed similarity-selection module (+Sim) and difference-selection module (++Diff), thereby quantifying their respective effects. All ablations are performed at token-retention ratios of 30% and 50%, with the results summarized in Table 5.

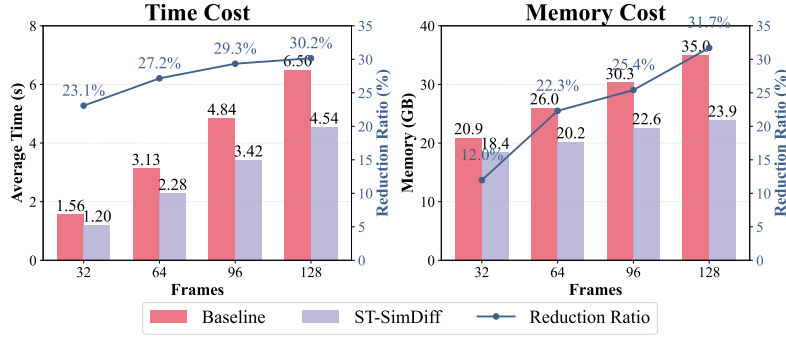


Figure 3: Computational-cost comparison between ST-SimDiff and the baseline LLaVA-Video model, measured by average inference time and peak GPU memory usage under a 30% token budget.

(1) We first evaluate the effectiveness of the +Sim module. The results show that introducing similarity modeling on top of the baseline consistently improves model performance. We further decompose the similarity strategy into three variants—spatial-only (Spatial), temporal-only (Temporal), and joint spatio-temporal modeling (Spa. + Tem.)—and observe that collaborative modeling of spatio-temporal associations among visual tokens provides the most effective redundancy-compression mechanism.

(2) After incorporating the optimal joint spatio-temporal similarity module, we introduce the difference-selection module (++Diff) to obtain the complete framework. This addition produces a substantial performance improvement, especially under the more aggressive 30% retention setting. The result illustrates the module’s complementary role: while +Sim primarily compresses redundancy, ++Diff functions as a safeguard that identifies and preserves distinctive event tokens defining the video’s narrative. By contrast, its marginal gain is smaller at a 50% retention ratio, because the less aggressive +Sim configuration is more likely to have already retained such event tokens and the overall model performance is closer to its upper bound. This ablation therefore supports our central motivation: the framework is most effective when it balances similarity-based redundancy compression with the preservation of difference-based events, particularly under demanding high-compression conditions.

5.5 COMPUTATIONAL COST ANALYSIS

To quantify the practical efficiency advantages of ST-SimDiff, we measure inference time and peak GPU memory usage under a 30% token budget. The comparison is conducted against the uncompressed baseline model (Baseline LLaVA-Video), and the results in Figure 3 cover video input lengths ranging from 32 to 128 frames.

Inference Time ST-SimDiff yields a substantial speedup. As the number of input frames increases, the baseline model’s average inference time rises sharply from 1.56 seconds at 32 frames to 6.50 seconds at 128 frames; in contrast, our method attenuates this growth and requires only 4.54 seconds on average for 128-frame inputs. Consequently, ST-SimDiff delivers increasingly pronounced time savings as videos become longer, improving from 23.0% at 32 frames to 30.2% at 128 frames.

Memory Cost ST-SimDiff also exhibits strong memory efficiency. The baseline model’s peak GPU memory usage increases approximately linearly with video length, rising from 20.9 GB at 32 frames to 35.0 GB at 128 frames. By substantially reducing the number of tokens forwarded to the large language model, our method markedly alleviates memory pressure: for 128-frame videos, ST-SimDiff uses only 23.9 GB of peak memory, corresponding to a 31.7% reduction relative to the baseline. This reduction enables longer videos to be processed on memory-constrained hardware and thereby broadens the method’s practical applicability.

In summary, ST-SimDiff not only achieves leading task performance, but also demonstrates clear computational advantages in both inference time and memory consumption, thereby confirming its efficiency for video understanding.

6 CONCLUSION

In this paper, we address the computational inefficiency encountered by Multimodal Large Language Models when processing long videos, a difficulty arising from excessive visual-token counts and from the insufficient preservation of critical content changes. We propose ST-SimDiff, a training-free framework that constructs a spatio-temporal graph to model complex associations among tokens. The method employs a dual-selection strategy in which similarity-based selection identifies redundancy through community detection, while temporal-difference-based selection preserves tokens located at key transition points. Extensive experiments on VideoMME and LongVideoBench show that ST-SimDiff substantially reduces processing time and memory usage while consistently achieving state-of-the-art performance. By balancing content representativeness with transitional event preservation, this work establishes a promising direction for constructing more efficient representations of video data.

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A VISUALIZATION OF THE TOKEN SELECTION PROCESS

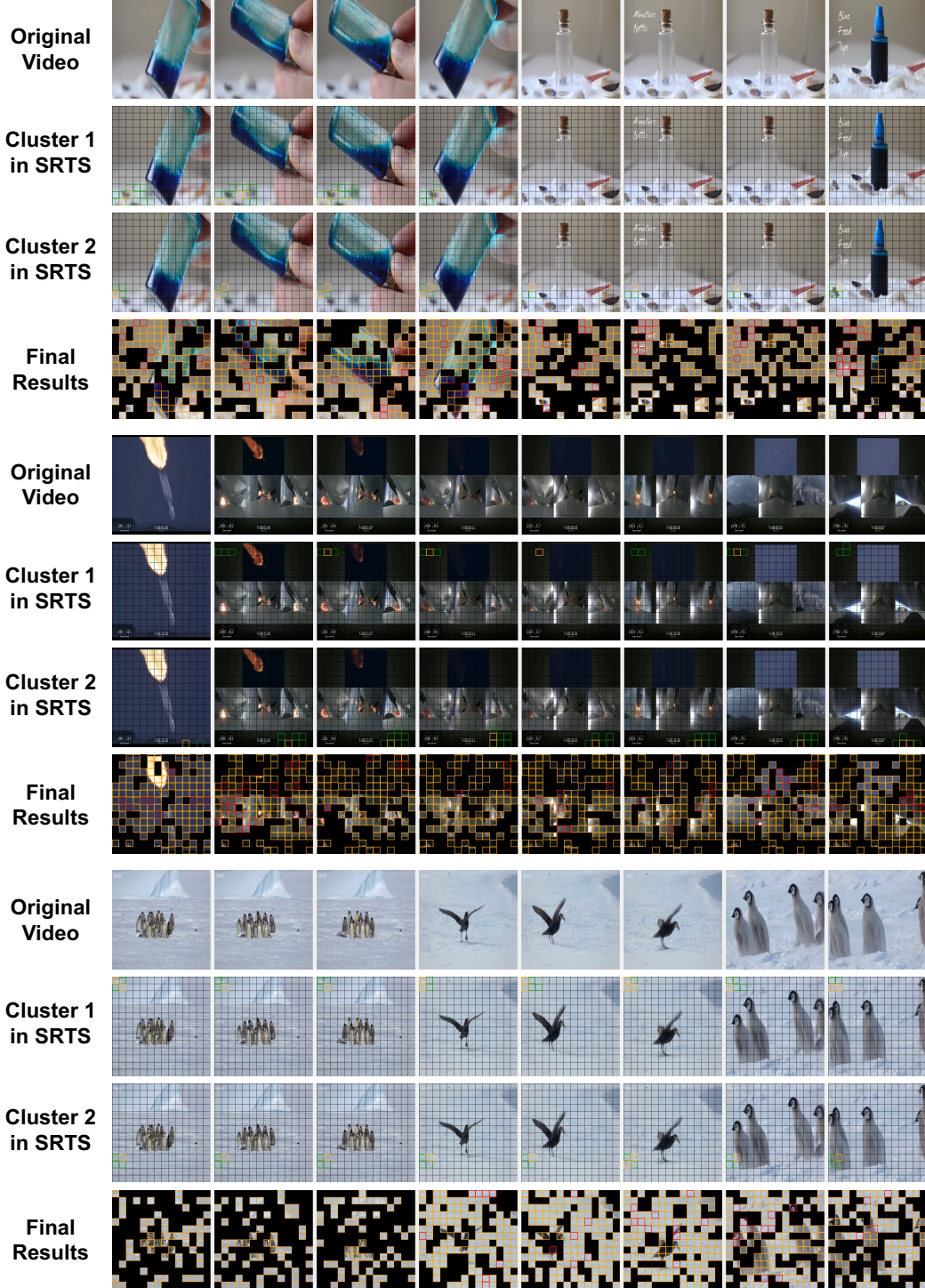


Figure 4: Visualization of the ST-SimDiff process, decomposing token selection into SRTS-detected communities (shown as grids in Cluster 1 and Cluster 2) and DETS-detected event-driven tokens. The final result illustrates the complementarity between sparse representative tokens (yellow), which encode stable content, and dense event tokens (red), which preserve dynamic actions.

To provide a more intuitive account of the ST-SimDiff framework, we present visualization examples of both intermediate and final token-selection results in Figure 4. The figure depicts the procedure on a sample video sequence involving dynamic object manipulation. For Similarity-based Representative Token Selection (SRTS), the rows labeled “Cluster 1” and “Cluster 2” illustrate the operation of the graph community detection algorithm: spatially and temporally redundant regions, including the static background and stationary objects, are organized into coherent communities. This visualization indicates that SRTS identifies stable visual elements and compresses them by retaining only a small number of central “representative tokens,” which appear as sparse yellow boxes in the final results. Through this mechanism, the model can efficiently accommodate large amounts of background redundancy.

Complementarily, Difference-based Event Token Selection (DETS) is especially important during dynamic intervals. As shown in the “Final Results” row, tokens activated by high temporal difference are marked with dense red bounding boxes. These red boxes align closely with the moving hand and the shifting blue liquid, confirming that the method localizes “turning points” at which the temporal difference exceeds the threshold (τ_{diff}). This selection behavior preserves fine-grained action details and rapid visual changes, enabling the model to capture the essential event structure without being dominated by static content.

Finally, the overlay in the final results makes the synergy between the two components visually explicit. The sparse yellow tokens summarize the stable background, whereas the dense red tokens concentrate on rapid motion, and this contrast provides direct evidence for our central motivation. ST-SimDiff mitigates the blind spots of similarity-only approaches by sampling densely around rapid changes, while simultaneously pruning spatiotemporal redundancies that importance-based methods may leave unresolved. Consequently, the framework achieves a favorable balance between computational efficiency and the preservation of critical visual information.

B COMPARISON RESULTS ON QWEN2.5-VL

Method	VideoMME				LongVideoBench			EgoSchema
	Short	Medium	Long	Overall	Relation	Perception	Overall	
Upper Bound (Full Performance)								
Qwen2.5-VL-64	74.2	62.7	51.8	62.9	54.2	64.8	59.2	63.0
Qwen2.5-VL-128	77.4	65.8	56.1	66.4	56.2	65.1	60.4	63.9
Token Retain Ratio $r = 30\%$								
ST-SimDiff-64	73.8	61.7	51.8	62.4	52.2	62.1	56.8	62.7
ST-SimDiff-128	75.4	64.7	55.2	65.1	53.1	65.0	58.6	64.4
Token Retain Ratio $r = 50\%$								
ST-SimDiff-64	74.2	62.1	52.2	62.9	54.2	63.2	58.4	62.7
ST-SimDiff-128	76.8	66.3	55.8	66.3	54.9	65.3	59.8	64.5

Table 4: Performance comparison on long-form video-understanding benchmarks using Qwen2.5-VL-7B with 64 and 128 input frames (%).

To further examine the generalization capability of the proposed framework, we conduct additional experiments with the recently released Qwen2.5-VL model. These evaluations, performed on VideoMME and LongVideoBench, are intended to determine whether the efficacy of ST-SimDiff extends beyond a single model architecture and remains applicable across different LVLM backbones.

The empirical results in Table 4 demonstrate that ST-SimDiff retains strong performance on Qwen2.5-VL. At a 30% token-retention ratio, the method achieves scores of 62.4 on VideoMME and 56.8 on LongVideoBench; when the retention ratio is increased to 50%, performance rises to 62.9 on VideoMME and 58.4 on LongVideoBench. Notably, under the 50% setting, VideoMME performance matches the upper-bound result obtained with all tokens, while the LongVideoBench score remains highly comparable to its full-token counterpart. These findings further support the cross-backbone generalization ability of the framework.

C ABLATION STUDY RESULTS ON NVILA

Method	VideoMME	LongVideoBench	EgoSchema	
Token Retain Ratio $r = 30\%$				
Baseline	59.3	53.8	50.2	
+ Sim	Spatial	59.5	54.5	50.9
	Temporal	59.6	54.3	51.2
	Spa. + Tem.	59.9	54.6	51.3
++ Diff (Proposed)	60.2	55.2	51.7	
Token Retain Ratio $r = 50\%$				
Baseline	60.9	55.3	51.5	
+ Sim	Spatial	61.1	55.7	51.9
	Temporal	61.3	55.6	51.8
	Spa. + Tem.	61.4	56.0	52.3
++ Diff (Proposed)	61.7	56.5	52.5	

Table 5: Component-wise ablation results on the NVILA-Video-8B model (%), measuring the effects of similarity selection and difference selection.

To further validate whether each component of the framework generalizes across base models, we conduct an additional set of ablation studies on NVILA-Video-8B (Liu et al., 2024c). The experimental setup follows the main-paper ablation: beginning with a Baseline strategy that contains only fundamental importance-based pruning, we progressively introduce the similarity-selection module (+ Sim) and the difference-selection module (++ Diff).

As shown in Table 5, the results again confirm the effectiveness of the framework design, with trends that are highly consistent with those observed on LLaVA-Video-7B. First, introducing the similarity module (+ Sim) on top of the baseline yields steady improvements at both 30% and 50% token-retention ratios. Within the similarity module, we compare spatial-only (Spatial), temporal-only (Temporal), and joint spatio-temporal (Spa. + Tem.) variants, and the joint strategy is consistently the most effective. For example, when $r = 50\%$, spatio-temporal joint modeling achieves 61.4% on VideoMME, outperforming the spatial-only and temporal-only alternatives; this result again highlights the value of jointly considering spatio-temporal associations when identifying redundant information. After integrating the optimal joint spatio-temporal similarity module, we further introduce the dissimilarity-selection module (++ Diff) to form the complete ST-SimDiff framework. The table shows that this addition yields the most substantial performance gains, thereby demonstrating the effectiveness and cross-model generalization of the two core modules—similarity selection and dissimilarity selection—and further validating the central principle of simultaneously handling redundancy and change.

D ABLATION RESULTS OF τ_{sim} AND τ_{diff} ON LLaVA-VIDEO

In our framework, the similarity threshold τ_{sim} and the difference threshold τ_{diff} are two key hyperparameters that respectively govern the selection of representative tokens and transitional tokens. To examine their effects, we conduct a series of ablation studies, with the results reported in Figure 6. The experiments indicate that both parameters produce a similar performance pattern: accuracy initially increases and then declines, while remaining relatively robust within a moderate range. For τ_{sim} , an overly low value results in imprecise community detection, whereas an overly high value can fragment semantic clusters and undermine their integrity. For τ_{diff} , an overly low value may introduce noise through excessive sensitivity, while an overly high value can cause the model to miss key events. Based on these results, the model achieves its best performance at $\tau_{sim} = 0.8$ and $\tau_{diff} = 0.2$, and we therefore adopt this configuration for all other experiments.

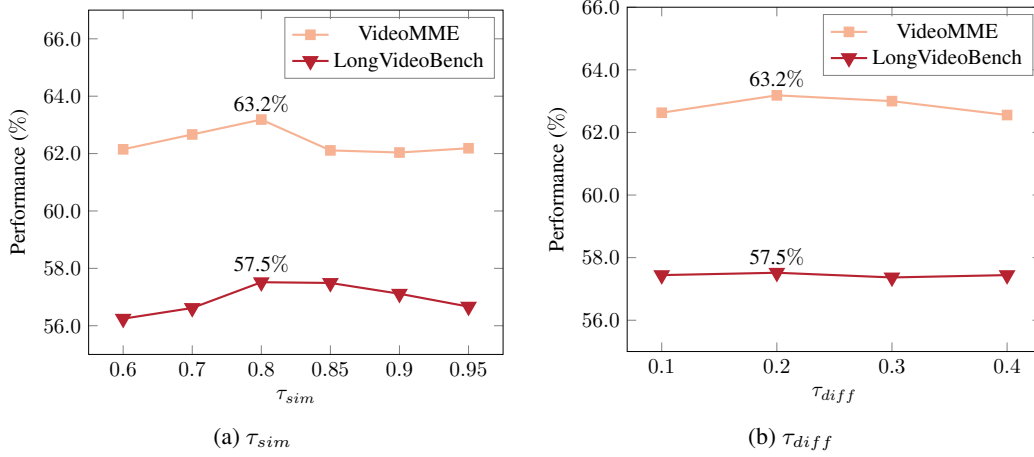


Figure 5: Ablation results for different values of τ_{diff} and τ_{sim} on VideoMME and LongVideoBench.

E ABLATION RESULTS OF τ_{sim} AND τ_{diff} ON NVILA-VIDEO

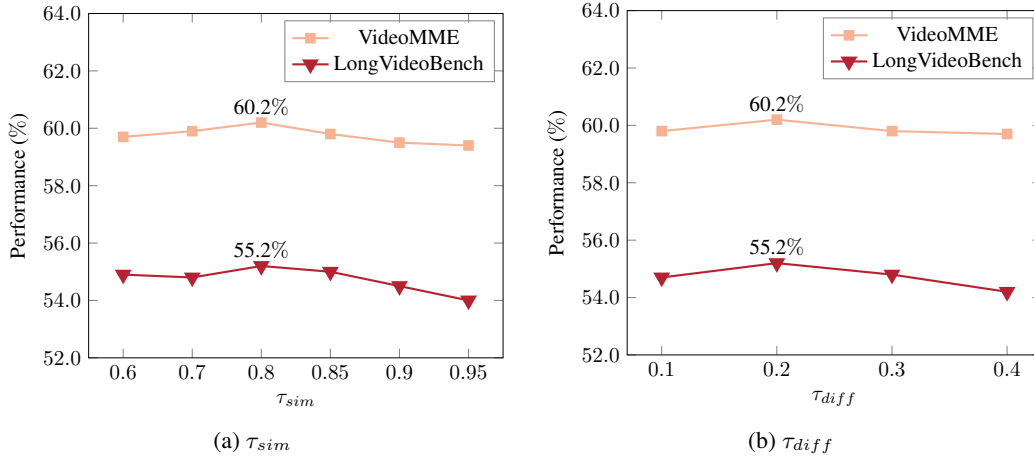


Figure 6: Ablation results for different values of τ_{diff} and τ_{sim} on NVILA-Video-8B.

To assess the robustness of the method with respect to its key hyperparameters, we conduct additional ablation studies on the NVILA-Video-8B model. As shown in Fig. 6, we evaluate how the similarity threshold, τ_{sim} , and the difference threshold, τ_{diff} , affect performance on VideoMME and LongVideoBench.

Impact of Similarity Threshold τ_{sim} . We test τ_{sim} across the range $[0.6, 0.95]$. The results indicate that model performance is relatively insensitive to moderate changes in this threshold. On VideoMME, the peak performance of 60.2% is achieved at $\tau_{sim} = 0.8$; similarly, on LongVideoBench, the optimal performance is also obtained at $\tau_{sim} = 0.8$, reaching 55.2%. Although performance decreases slightly when the threshold becomes either too high or too low, the overall fluctuation remains limited, confirming that the method is stable across a broad interval of similarity-threshold values.

Impact of Difference Threshold τ_{diff} . We test τ_{diff} over the range $[0.1, 0.4]$. As with τ_{sim} , performance remains robust to variations in τ_{diff} . On both benchmarks, the optimal result occurs around $\tau_{diff} = 0.2$, with VideoMME reaching 60.2% and LongVideoBench reaching 55.2%. This pattern

suggests that a difference threshold near 0.2 is most effective for capturing key events while maintaining token compression.

In summary, these ablation studies confirm that the proposed method is robust with respect to its principal hyperparameters. The results justify the use of $\tau_{\text{sim}} = 0.8$ and $\tau_{\text{diff}} = 0.2$ in the main experiments, since these values consistently yield optimal performance across different benchmarks and base models.